

FLOATING GROUNDS

RETHINKING RESILIENCE IN JAKARTA



TYPE: Computational Design

YEAR: 2025

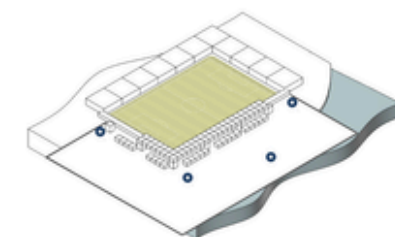
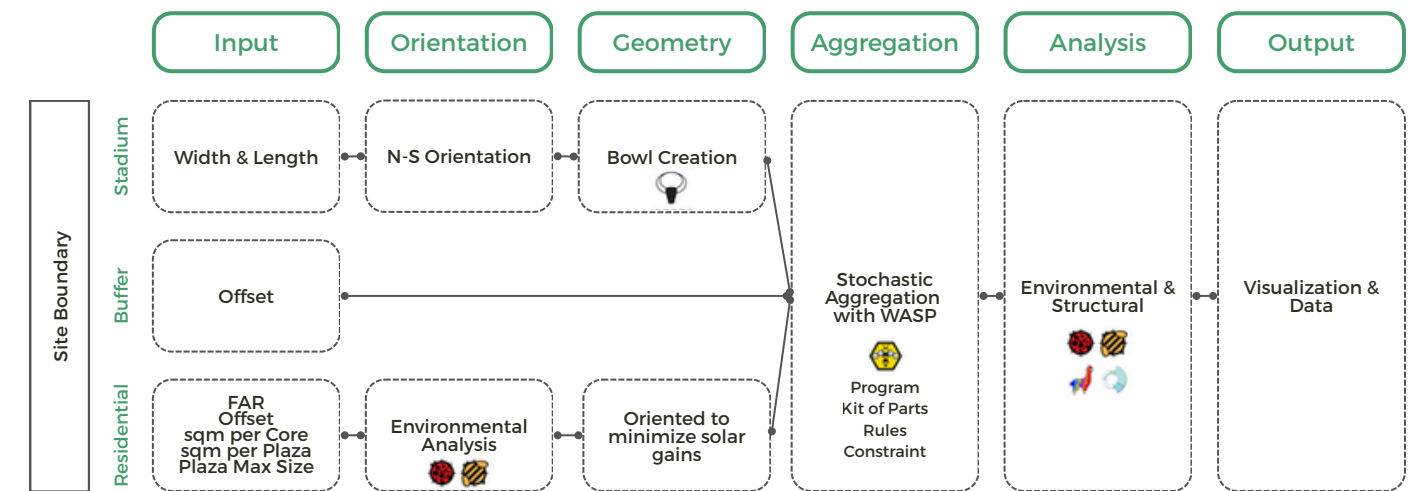
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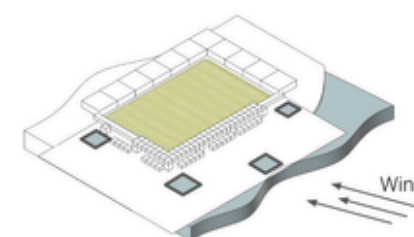
CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

“Floating Grounds” addresses Jakarta's land subsidence and flooding through a modular, water-based urban design. The project proposes a resilient floating stadium and residential hub in the Penjaringan District, merging mega-projects with local community needs. Utilizing lightweight timber and floating pontoons, the design minimizes carbon footprint while ensuring stability via Archimedes’ principle. The project follows a parametric and data-driven workflow ensuring a sustainable, scalable solution for flood-prone cities.

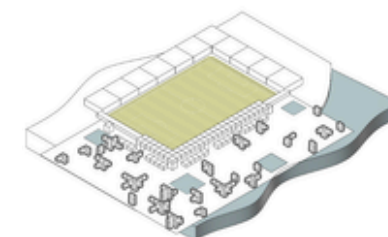
WORKFLOW



Floating platform & start points



Opening the ecosystem plazas



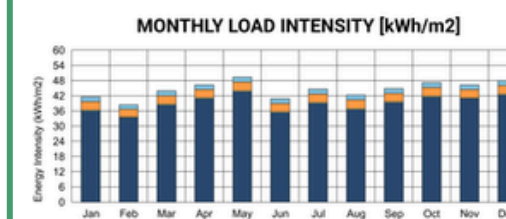
Creating core and circulations



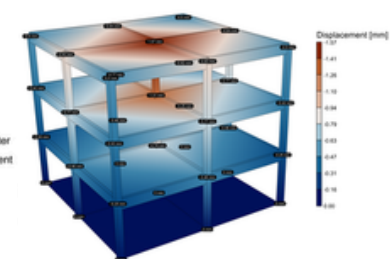
Run aggregation

STRUCTURAL & ENVIRONMENTAL ANALYSIS

Orientation and window ratio optimization reduced solar gains for residential units. Carbon footprint guided material choice, while structural analysis of the timber frame sized the floating structure.



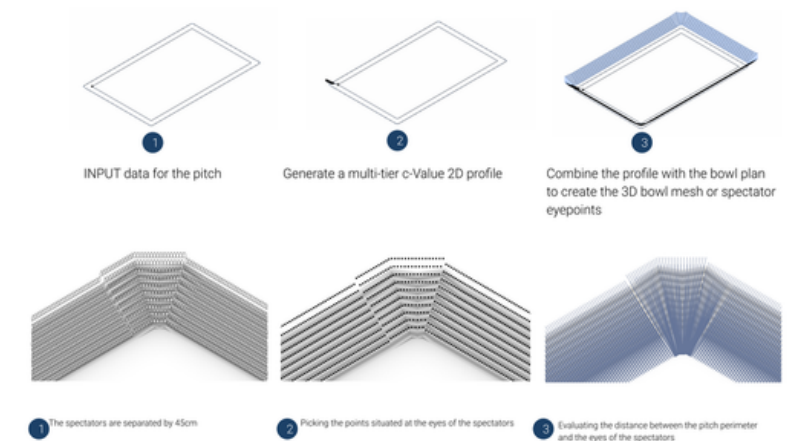
Energy Consumption



Max Deflection

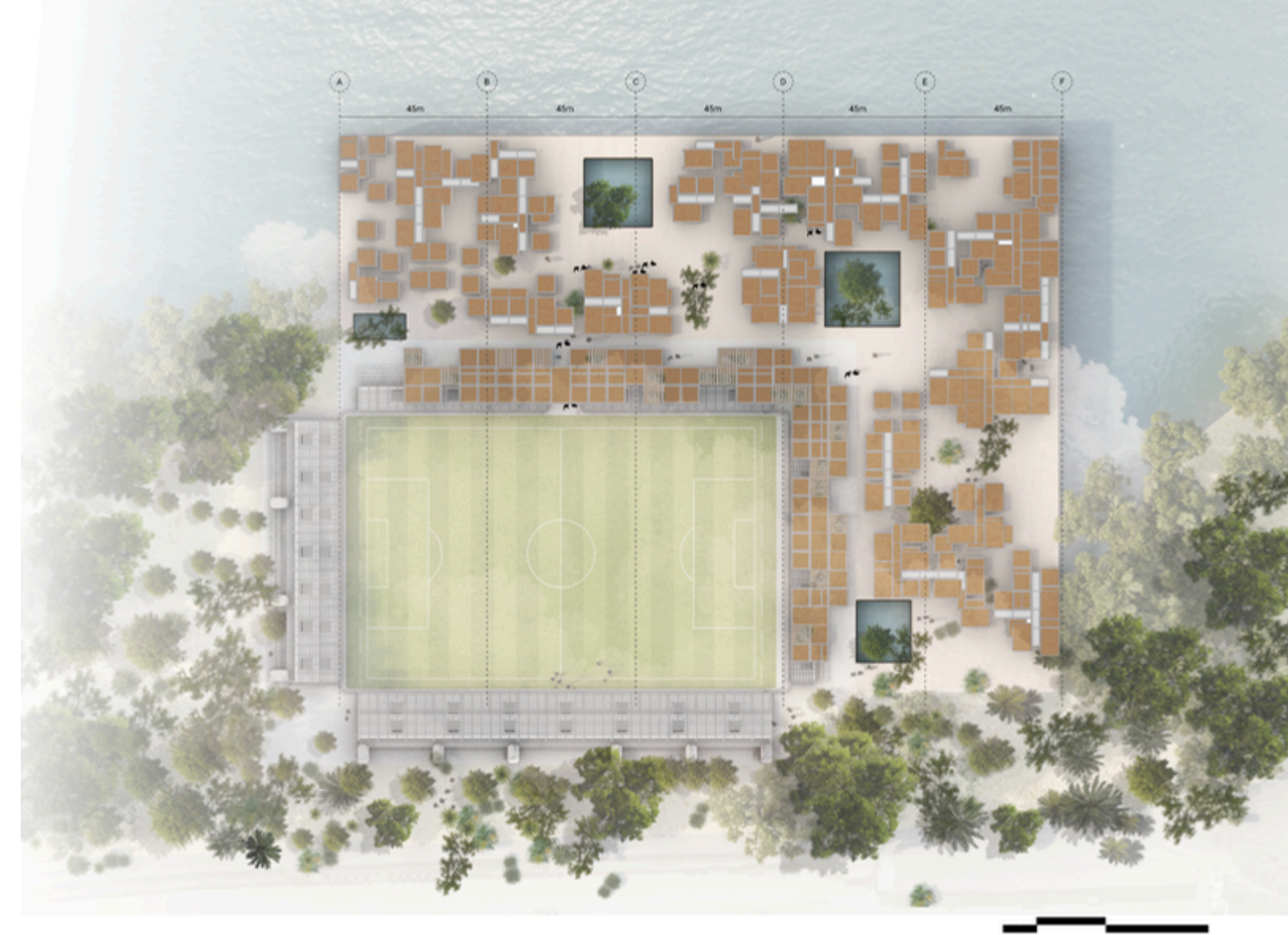
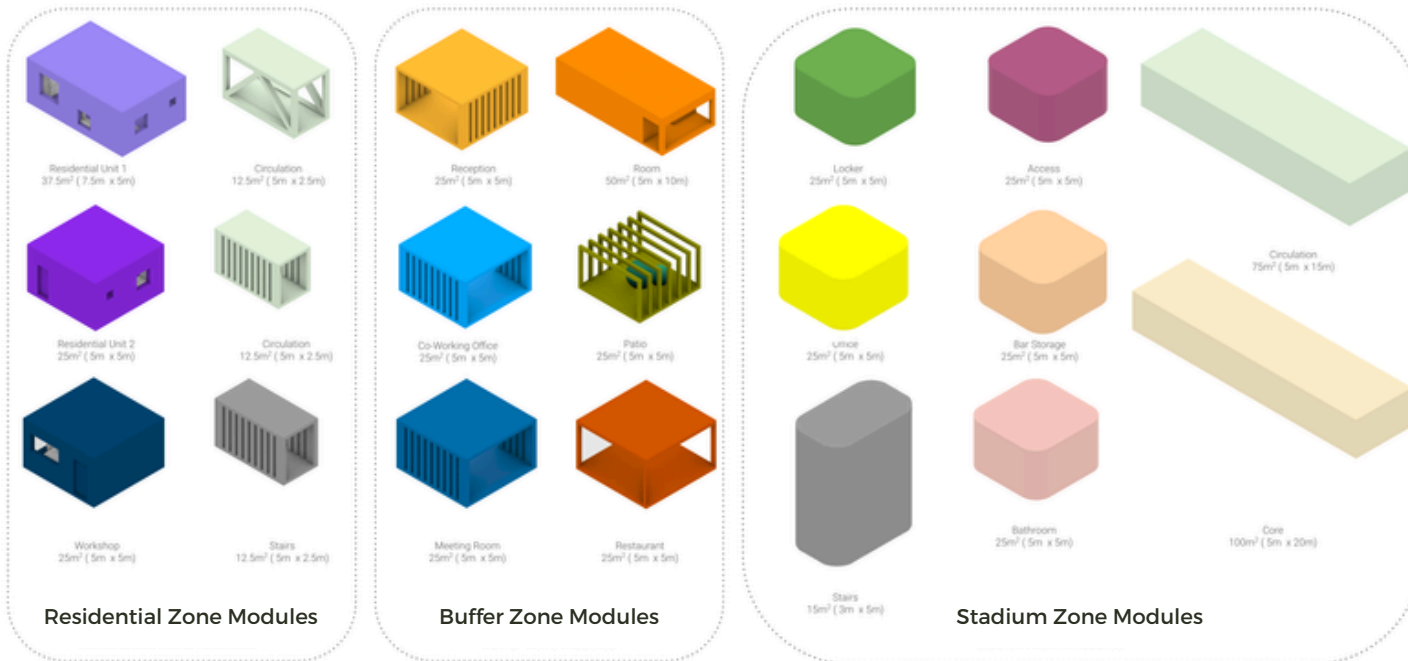
BOWL CREATION

The bowl's parametrization uses the C-value method, via the Toro plug-in, to ensure unobstructed views for any pitch dimension.



MODULAR UNIT AGGREGATION

All programs are built from a standardized modular system based on a 5x5 m grid. Modules are aggregated through rules-based logic that controls adjacency, access, vertical circulation and open space creation.



TESTING

The workflow allows rapid iteration, performance testing and adaptation to different sites while maintaining spatial and structural coherence.

